



Databases and its Applications

Advanced NoSQL and Emerging Database trends

Pooja T S

Computer Applications

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Replication and Replica Sets in MongoDB

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Computer Applications



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Replication in MongoDB – Motivation

► Why replication is required

A single database server is a single point of failure
Hardware failure or crashes cause application downtime
Replication ensures availability and fault tolerance

► What replication provides

Multiple copies of the same data on different machines
Automatic failover when a server goes down
Improved read availability (read from secondary nodes)



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Replication – Core Idea

► Basic concept

Data written once is copied to multiple servers

One server accepts writes, others replicate changes

All replicas maintain the same dataset

► MongoDB approach

MongoDB uses **Replica Sets** for replication

Replica sets are built-in (not an add-on feature)

Replication is transparent to applications



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What is MongoDB Replication?

► Definition

MongoDB replication is the process of synchronising data across multiple servers

All participating servers maintain identical copies of data

Replication is implemented using **Replica Sets**

► Key objective

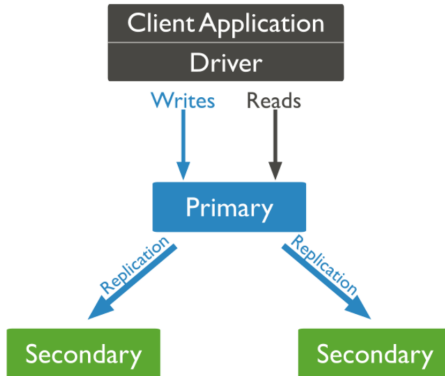
Ensure high availability of data

Provide fault tolerance without manual intervention



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MongoDB Replication



Basic replication methodology



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What is a Replica Set?

► Definition

A replica set is a group of MongoDB servers

All members store the same data

One node is primary, others are secondary

► Why replica sets are preferred

Automatic failover without manual intervention

Data consistency across nodes

High availability by design



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Replica Set – Core Concept

► Replica Set

A replica set is a group of MongoDB servers that maintain the same data

Each server in the set runs a `mongod` process

Data is automatically replicated among members

► Logical organisation

One node acts as **Primary**

Remaining nodes act as **Secondary**



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Primary Node in a Replica Set

► **Primary node responsibilities**

Handles all write operations

Records changes to the dataset

Propagates updates to secondary nodes

► **Primary uniqueness**

At any time, only one primary exists in a replica set

Guarantees consistency of write operations



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Secondary Nodes in a Replica Set

▶ **Secondary node characteristics**

- Replicate data from the primary node
- Maintain complete copies of the dataset
- Do not accept direct write operations

▶ **Read behaviour**

- Secondary nodes can serve read requests
- Improves read scalability and availability



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Replica Set Members – Roles

► **Primary node**

Handles all write operations

Updates are recorded and replicated

Only one primary exists at a time

► **Secondary nodes**

Replicate data from the primary

Can serve read requests (optional)

Promote themselves during failover



Data Synchronisation in Replica Sets

► **How replication works**

Primary node records all data changes

Secondary nodes continuously copy changes from the primary

Ensures data consistency across all members

► **Continuous process**

Synchronisation occurs automatically

No manual intervention required under normal conditions



Replication Mechanism – How Data Flows

► Operation log (oplog)

Primary records all write operations in an oplog

Oplog is an ordered list of data changes

Secondaries continuously read and apply oplog entries

► Result

All nodes eventually reach the same state

Replication is asynchronous by default

Small replication lag may exist



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Heartbeat Mechanism

► Purpose of heartbeats

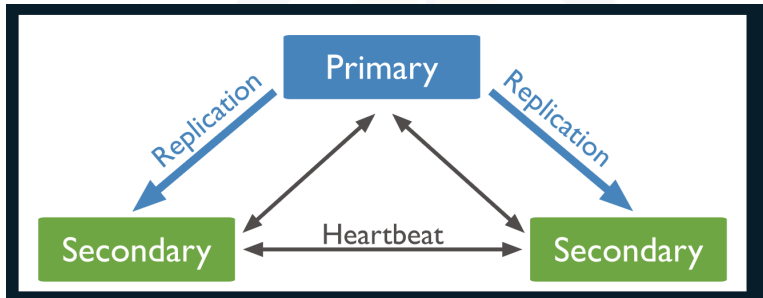
Nodes exchange heartbeat messages at regular intervals
Used to monitor availability of replica set members

► Failure detection

Missing heartbeats indicate node failure
Triggers election process if primary becomes unavailable



Heartbeat Mechanism





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Benefits of Replica Sets

► Key advantages

High availability

Automatic recovery from failures

Improved read scalability

► Production relevance

Essential for real-world MongoDB deployments

Forms the foundation for reliable NoSQL systems



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Automatic Failover in Replica Sets

► Failure detection

Nodes continuously send heartbeat messages

If primary stops responding, an election is triggered

Cluster agrees that primary is unavailable

► Election process

One secondary is elected as new primary

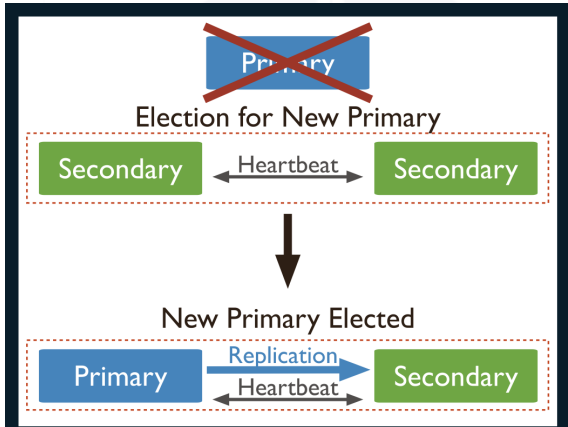
Election is automatic and fast

Applications reconnect transparently



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Automatic Failover in Replica Sets





Read and Write Behaviour in Replica Sets

► Write behaviour

Writes always go to the primary

Ensures a single authoritative write source

Prevents conflicting updates

► Read behaviour

Reads can go to primary or secondaries

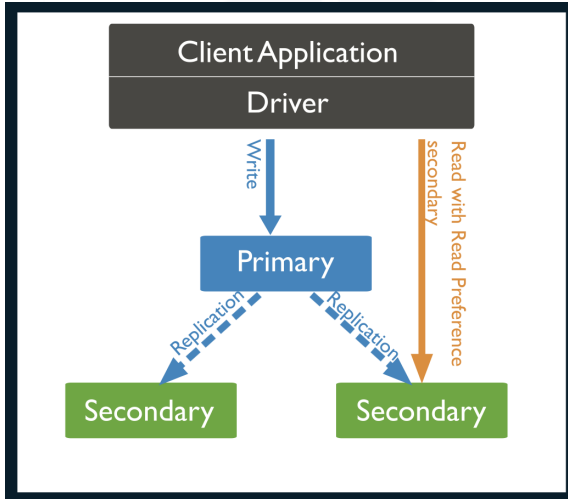
Controlled using read preferences

Helps distribute read load



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Read Behaviour in Replica Sets





► Replication

- Same data copied to multiple nodes
- Improves availability and fault tolerance
- Does not increase total data capacity

► Sharding

- Data is split across nodes
- Improves scalability and storage capacity
- Often combined with replication



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Common Replica Set Configurations

► **Typical setup**

- One primary and two secondaries
- Provides majority voting for elections
- Prevents split-brain scenarios

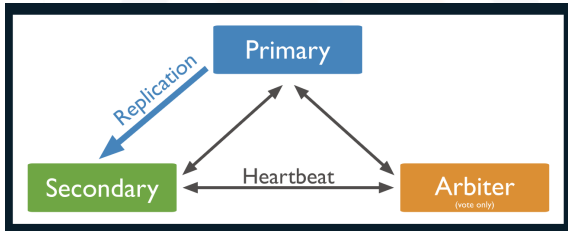
► **Arbiters (conceptual)**

- Participate in elections but do not store data
- Used when resource constraints exist
- Not recommended for critical data



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Common Replica set Configuration





Replication – Limitations and Considerations

► Limitations

Replication does not improve write throughput

Replication lag can affect read consistency

Network partitions can impact elections

► Design considerations

Always deploy odd number of voting members

Monitor replication lag

Combine with sharding for large-scale systems

